CHAPTER 11

CORRELATION AND REGRESSION

CHAPTER ESSENCE AT A GLANCE:
Master bivariate relationships: Karl Pearson's correlation coefficient, Spearman's rank correlation, regression equations of Y on X and X on Y, regression coefficients, and geometric properties.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Correlation vs Regression:** Correlation measures strength and direction of linear association between two variables (-1 <= r <= +1); Regression estimates the average value of a dependent variable from known values of an independent variable.

• **Karl Pearson Product Moment r:** Cov(X, Y) divided by [sigma$_X$ * sigma$_Y$]; independent of change of origin and scale; r = +1 indicates perfect positive correlation, r = -1 perfect negative, r = 0 zero linear correlation.

• **Spearman Rank Correlation (rho):** Ranks qualitative or ordinal data; rho = 1 - [6 * sum d^2] divided by [n(n^2 - 1)]; bounded strictly between -1 and +1.

• **Regression Lines and Coefficients:** Two regression lines: Y on X [Y - Y$_{bar}$ = b$_{yx}$(X - X$_{bar}$)] and X on Y [X - X$_{bar}$ = b$_{xy}$(Y - Y$_{bar}$)]; correlation coefficient is geometric mean of regression coefficients: r = +- sqrt(b$_{yx}$ * b$_{xy}$).

ESSENTIAL STATISTICAL FORMULAS & MEASURES

$$r = \frac{\sum(x-\bar{x})(y-\bar{y})}{\sqrt{\sum(x-\bar{x})^2 \sum(y-\bar{y})^2}} = \frac{\mathrm{Cov}(X,Y)}{\sigma_X \sigma_Y}$$

Karl Pearson Product-Moment Correlation Coefficient

$$\rho = 1 - \frac{6\sum d_i^2}{n(n^2-1)} \rightarrow -1 \leq \rho \leq +1$$

Spearman Rank Correlation for Non-Tied Ranks

$$b_{yx} = r\,\frac{\sigma_y}{\sigma_x} \quad \text{and} \quad b_{xy} = r\,\frac{\sigma_x}{\sigma_y}$$

Regression Slopes of Y on X and X on Y

$$r = \pm\sqrt{b_{yx} \times b_{xy}} \quad \text{and} \quad \frac{b_{yx}+b_{xy}}{2} \geq r$$

Geometric and Arithmetic Mean Properties of Regression Slopes

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Both regression coefficients b$_{yx}$ and b$_{xy}$ must have the SAME algebraic sign (both positive or both negative); correlation r takes the same sign as the coefficients.

• **Alert:** If one regression coefficient is greater than 1, the other must be less than 1 because their product r^2 cannot exceed 1.

• **Alert:** The two regression lines always intersect at the point of their arithmetic means: (X$_{bar}$, Y$_{bar}$).

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Distinguish between Correlation and Regression on the basis of: (a) Nature of relationship, (b) Symmetry, (c) Units of measurement. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **(a) Nature of Relationship:** Correlation measures the degree and direction of linear co-variation between variables; Regression establishes a functional mathematical relationship to predict dependent variable from independent variable. **[1 Mark]**

• **(b) Symmetry:** Correlation is symmetric ($r_{xy} = r_{yx}$); Regression is asymmetric (regression of Y on X is different from X on Y). **[1 Mark]**

• **(c) Units:** Correlation coefficient r is a pure dimensionless number; Regression coefficients are expressed in original physical measurement units. **[1 Mark]**

2. State any four important properties of regression coefficients. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **1. Same Algebraic Sign:** Both regression coefficients b_{yx} and b_{xy} and correlation r always have the same sign (all positive or all negative). **[1 Mark]**

• **2. Geometric Mean Property:** Correlation coefficient is the geometric mean of the two regression coefficients: r = +- square root of (b_{yx} * b_{xy}). **[1 Mark]**

• **3. Magnitude Bound:** If one regression coefficient is greater than unity (> 1), the other must be less than unity (< 1), since $r^2 \leq 1$. **[1 Mark]**

• **4. Arithmetic Mean Dominance:** The arithmetic mean of the two regression coefficients is greater than or equal to correlation r: (b_{yx} + b_{xy}) divided by 2 ≥ r. **[1 Mark]**

3. For two variables X and Y: Mean of X = 20, Mean of Y = 40, Variance of X = 9, Variance of Y = 25, and correlation coefficient r = 0.6. Find: (a) The two regression coefficients, (b) The regression equation of Y on X, (c) Estimate Y when X = 25. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Parameters:** $X_{bar} = 20$, $Y_{bar} = 40$, $sigma_x$ = square root of 9 = 3, $sigma_y$ = square root of 25 = 5, r = 0.6. **[1 Mark]**

• **(a) Regression Coefficients:**
• b_{yx} = r * ($sigma_y$ divided by $sigma_x$) = 0.6 * (5 divided by 3) = 1.0.
• b_{xy} = r * ($sigma_x$ divided by $sigma_y$) = 0.6 * (3 divided by 5) = 0.36. **[1 Mark]**

• **(b) Regression Equation of Y on X:** $Y - Y_{bar} = b_{yx} * (X - X_{bar})$. Y - 40 = 1.0 * (X - 20). Y = X + 20. **[1 Mark]**

• **(c) Estimate Y when X = 25:** Y = 25 + 20 = 45. **[1 Mark]**

4. The ranks of 5 competitors in two musical events were: Competitor A (Rank1=1, Rank2=2), B (2, 1), C (3, 4), D (4, 3), E (5, 5). Calculate Spearman's rank correlation coefficient. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Rank Differences d = R1 - R2:** -1, +1, -1, +1, 0. **[1 Mark]**

• **Squared Differences d^2:** 1, 1, 1, 1, 0. Sum of d^2 = 4. n = 5. **[1 Mark]**

• **Formula:** ρ = 1 - [6 * sum d^2] divided by [$n(n^2 - 1)$] = 1 - [6 × 4] divided by [5(25 - 1)] = 1 - 24 divided by 120 = 1 - 0.20 = +0.80. High positive correlation in rankings. **[1 Mark]**

5. At what point do the two lines of regression intersect? What does it imply if the two lines coincide? [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Intersection Point:** The two regression lines always intersect at the point of their arithmetic means: (X_{bar}, Y_{bar}). **[1 Mark]**

• **Implication of Coincidence:** If the two regression lines coincide, correlation is perfect: r = +1 or r = -1. If they are perpendicular, r = 0. **[1 Mark]**

6. Can b_{yx} = -1.2 and b_{xy} = +0.6 be the regression coefficients of a bivariate distribution? Justify. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Conclusion:** No, this is mathematically impossible. **[1 Mark]**

• **Justification:** Both regression coefficients must always have the SAME algebraic sign because $r = +- \sqrt{(b_{yx} * b_{xy})}$. If they have opposite signs, their product is negative and square root yields an imaginary number. **[1 Mark]**